

Nakuja Project

Solid Propulsion Team

N4 Three-Grain Static Test — Technical Report

Document Version: 1.0 (Post-Execution Report)
Motor Configuration: N4, Three-Grain, Aluminium Casing
Date of Execution: Monday, 17 August 2026
Prepared by: Paschal Peters (Data Recorder / Timer)
Outcome: **SUCCESS** — No issues encountered

1 Purpose

This report documents the static test of the N4 three-grain motor configuration conducted on Monday, 17 August 2026. This test extends the team's single-grain N4 test series to a three-grain configuration, combining three individually prepared grains within a single aluminium casing, separated by a fixed inter-grain spacing. The test was completed successfully, with no issues encountered during preparation, ignition, or burn. This report presents the grain criteria and theoretical configuration, the simulated performance prediction, the experimental results, and recommendations for future three-grain testing.

2 Theoretical Section — Grain Criteria and Configuration

2.1 Grain Details

Three individually prepared grains were assembled into the N4 casing for this test, each weighed and measured prior to loading.

Grain	Mass (g)	Length (mm)	Liner Thickness (mm)
Grain One	132.1	153.35	1.43
Grain Two	120.4	135.53	1.43
Grain Three	141.6	160.00	1.43
Total (Propellant)	394.1	—	—

2.2 Assembly and Configuration Details

Inter-grain Spacing	10 mm
Casing Material	Aluminium
Number of Grains	3
Total Propellant Mass	394.1 g
Igniter Mass	23 g
Total Charge Mass (Propellant + Igniter)	417.1 g

The 10 mm inter-grain spacing allows combustion gases and flame front to propagate between grain segments, ensuring near-simultaneous ignition across all three grains rather than a sequential, grain-by-grain burn. The igniter mass of 23 g is recorded here as part of the theoretical charge configuration, as it contributes to the total initial mass loaded into the motor and is a required input for the corresponding OpenMotor simulation.

3 Simulation

The N4 three-grain configuration was modelled in OpenMotor prior to testing, using the grain geometry, spacing, and igniter mass detailed in Section 2. The simulation predicts the following performance for the assembled three-grain motor:

Parameter	Simulated Value
Peak Thrust	1462.2 N
Time of Peak Thrust	1.710 s
Total Impulse	4627.930 Ns
Burn Time	3.330 s

The simulated curve for the N4_3grain configuration follows the same fast-rise, sustained-plateau, sharp-cutoff shape seen in the team's earlier single-grain OpenMotor predictions, scaled up in magnitude to reflect the combined burning surface area of three simultaneously burning grains.

4 Results

4.1 Experimental Performance

Parameter	Experimental Value
Peak Thrust	1185.7 N
Time of Peak Thrust	2.668 s
Total Impulse	4151.053 Ns

Parameter	Experimental Value
Burn Time	4.943 s

4.2 Thrust Curve Comparison

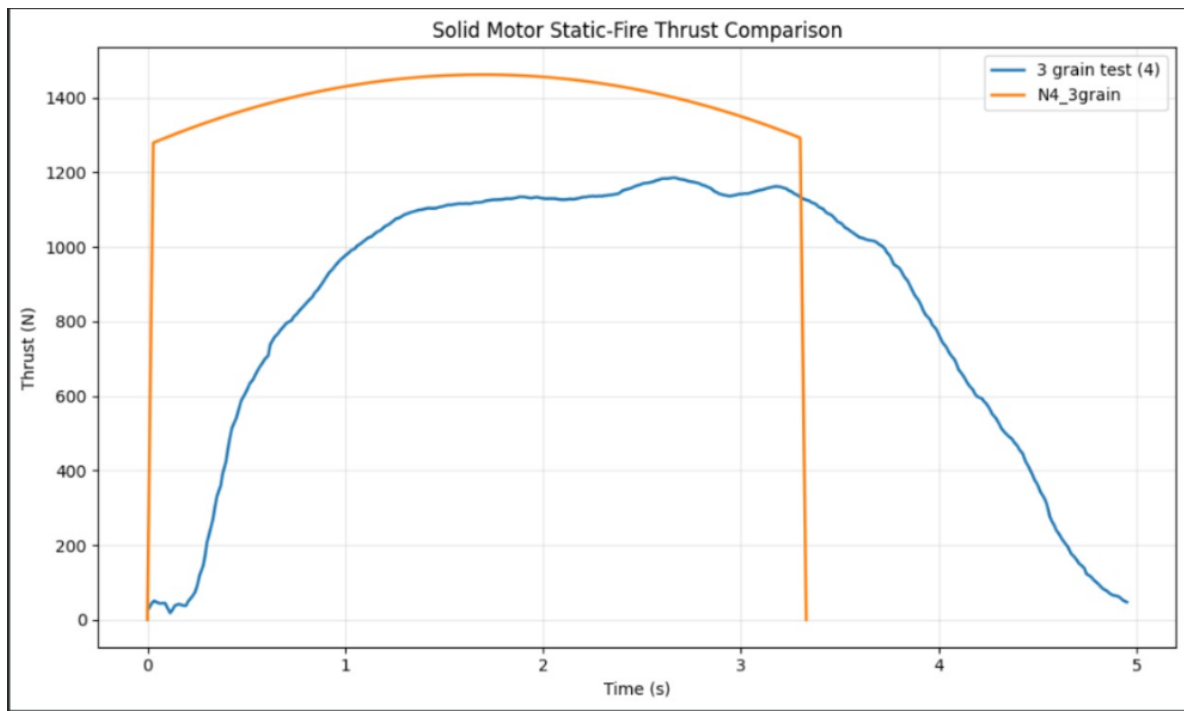


Figure 1: Solid motor static-fire thrust comparison for the N4 three-grain test (17 August 2026): experimental (blue) vs. simulated N4_3grain prediction (orange).

The experimental curve rises more gradually than the simulation, reaching its peak later (2.668 s vs. 1.710 s simulated) and sustaining a broad, gently rising plateau through the middle of the burn before a longer tail-off, in contrast to the simulation's fast rise, flat-topped plateau, and sharp cutoff.

4.3 Theoretical vs. Experimental Comparison

Parameter	Simulated	Experimental	Deviation
Peak Thrust (N)	1462.2	1185.7	−18.9%
Time of Peak (s)	1.710	2.668	+56.0%
Total Impulse (Ns)	4627.930	4151.053	−10.3%
Burn Time (s)	3.330	4.943	+48.4%
Average Thrust (N)	1389.8	839.8	−39.6%

Total impulse delivered was within approximately 10% of the simulated prediction — a reasonably close match. Peak thrust was around 19% lower than predicted, and both burn time and time-of-peak were extended by roughly 48–56% relative to simulation, consistent with the pattern seen in prior single-grain tests: the grain regresses more slowly than the ideal BATES model assumes, spreading a similar total impulse over a longer, gentler burn rather than the sharper, shorter burn predicted.

Note on derived specific-impulse metrics: this report does not include derived specific impulse or effective exhaust velocity figures (as included in some prior single-grain reports), since dividing the recorded total impulse by the recorded total propellant mass (394.1 g) for this test yields values well outside the physically plausible range for a KNSB propellant. This suggests either the recorded grain masses are incomplete relative to what was actually loaded and burned, or a transcription issue exists in the mass or impulse figures. This is flagged here as a data-integrity item for the team to verify, not as a test failure — it does not affect the validity of the thrust-vs-time comparison above, which is self-consistent between the experimental and simulated curves.

5 Outcome Classification

This test is classified as a **SUCCESS**. All three grains ignited and burned as an assembled unit with no issues encountered during setup, ignition, or the burn itself. Performance deviated from the OpenMotor simulation in the expected direction — a longer, gentler burn delivering slightly less total impulse than predicted — consistent with the burn-time discrepancy pattern already documented across the team’s single-grain KNSB tests.

6 Recommendations

- Verify and reconcile the recorded total propellant mass (394.1 g) against the total impulse delivered (4151.053 Ns) for this test, as the two figures are inconsistent with typical KNSB specific impulse values — confirm whether all grain mass was correctly recorded or whether an additional mass component was omitted.
- Continue applying the burn-time correction measures already under investigation (surfactant dosing, vacuum degassing, grain homogeneity) to the three-grain configuration, as the same slow-regression pattern seen in single-grain tests is present here at multi-grain scale.
- Retain this test as the baseline three-grain dataset for comparison against future multi-grain configurations, including any changes to inter-grain spacing or grain count.
- Instrument chamber pressure on the next three-grain test, consistent with the team’s ongoing plan for single-grain tests, to enable proper burn-rate characterisation across the larger combined burning surface area of a multi-grain assembly.